

Timothy Trujillo

Research-focused curriculum vitae | September 2026

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Research interests

Mathematical logic, set theory, ultrafilters, topological Ramsey spaces, nonstandard models of arithmetic, and mathematical foundations of collective choice. Current proposed research concerns infinite ethics and universal consequentialist systems with compatible aggregation and evaluation.

Academic appointments

Sam Houston State University, Mathematics and Statistics

Associate Professor of Mathematics, September 2023-August 2026. Promoted with tenure effective September 1, 2023; faculty resignation effective August 31, 2026.

Assistant Professor of Mathematics, 2017-2023.

Colorado School of Mines

Postdoctoral Teaching Scholar in Mathematics, 2015-2017.

Texas State University

Lecturer in Mathematics, 2014-2015.

University of Denver

Instructor and Graduate Teaching Assistant in Mathematics, 2008-2014.

New Mexico Institute of Mining and Technology

Instructor in Mathematics, 2006-2008.

Pending research affiliation

Sam Houston State University: non-compensated Research Affiliate, intended to begin September 1, 2026; administrative processing pending as of September 8, 2026. Exact appointment title and activation remain to be confirmed.

Education

Ph.D. in Mathematics, University of Denver, 2014. Advisor: Natasha Dobrinen. Dissertation: Topological Ramsey spaces, associated ultrafilters and their applications to the Tukey theory of ultrafilters and Dedekind cuts of nonstandard arithmetic.

M.S. in Mathematics, New Mexico Institute of Mining and Technology, 2008. Specialization in analysis. Advisor: Bixiang Wang. Thesis: Attractors for nonlinear reaction-diffusion equations in unbounded domains.

B.S. in Mathematics, New Mexico Institute of Mining and Technology, 2006. Minor in education.

Selected publications

Timothy Trujillo. Parametrizing by the Ellentuck Space. *Fundamenta Mathematicae* 255 (2021). [Publisher record](#)

Timothy Trujillo. Condorcet's Paradox and Ultrafilters. *The American Mathematical Monthly* 127(2) (2020), 169-174. DOI: [10.1080/00029890.2020.1680200](https://doi.org/10.1080/00029890.2020.1680200)

Timothy Trujillo, Connor Mattes, J. Menard, and Zachary Chaney. Triangular Ramsey numbers. *INTEGERS* 19 (2019). [Article PDF](#)

Natasha Dobrinen, Jose Mijares, and Timothy Trujillo. Topological Ramsey spaces from Fraisse classes, Ramsey-classification theorems, and initial structures in the Tukey types of p-points. *Archive for Mathematical Logic* (2017). DOI: [10.1007/s00153-017-0540-0](https://doi.org/10.1007/s00153-017-0540-0)

Timothy Trujillo. Selective but not Ramsey. *Topology and its Applications* 202 (2016). DOI: [10.1016/j.topol.2015.12.088](https://doi.org/10.1016/j.topol.2015.12.088)

Timothy Trujillo. Ramsey for R1 ultrafilter mappings and their Dedekind cuts. *Mathematical Logic Quarterly* 61 (2015). DOI: [10.1002/malq.201300078](https://doi.org/10.1002/malq.201300078)

Timothy Trujillo and Bixiang Wang. Continuity of strong solutions of the reaction-diffusion equation in initial data. *Nonlinear Analysis: Theory, Methods & Applications* 69 (2008). DOI: [10.1016/j.na.2007.08.032](https://doi.org/10.1016/j.na.2007.08.032)

Research support and current project

SHSU Enhanced Research Grant, Infinite ethics: a non-Archimedean approach. The grant-supported work led to Condorcet's Paradox and Ultrafilters. The current proposal, Infinite Ethics: Toward a Universal Consequentialist System, develops the broader program of context-sensitive composition, non-Archimedean evaluation, and coherent extension of partial systems. Universal existence and extension theorems are research objectives.

Selected mentoring, teaching, and service

Directed undergraduate and graduate studies in mathematical logic, set theory, and social choice; supervised student work leading to conference presentations and publications. Co-advised Colorado School of Mines Putnam teams in 2015 and 2016. Taught undergraduate mathematics and graduate mathematical logic at SHSU, with course development in proof-based mathematics and mastery-oriented assessment.

Service at SHSU included chairing a hiring committee for two tenure-track mathematics positions, department policy work, and co-organizing the logic seminar. Project NExT Fellow, Mathematical Association of America, 2017; Texas Section NeXT Fellow, 2017-2018.